

Lab 13: Code Refactoring – Improving Legacy Code with AI Suggestions

Name: Syed Murtaza

Hall ticket no: 2303A51259

Batch no: 19

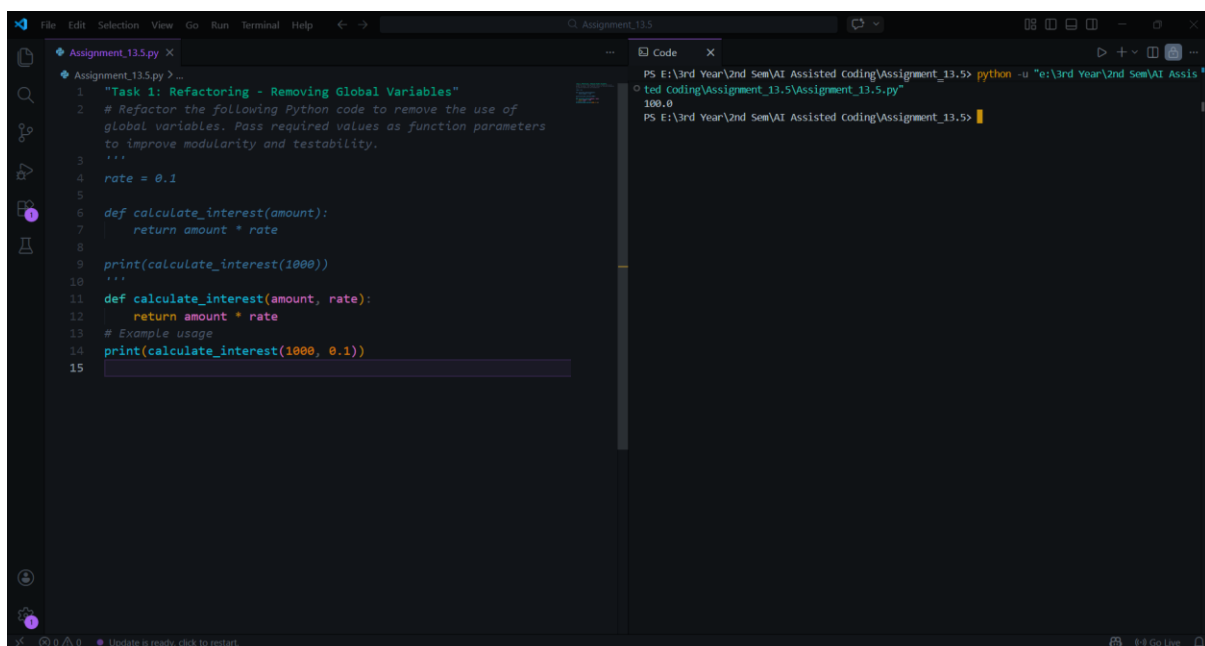
Task 1: Refactoring – Removing Global Variables

Prompt:

Refactor the following Python code to remove the use of global variables. Pass required values as function parameters to improve modularity and testability.

```
rate = 0.1
def calculate_interest(amount):
    return amount * rate
print(calculate_interest(1000))
```

Code & Output:

The screenshot shows a code editor with two panes. The left pane displays the refactored Python code for 'Assignment_13.5.py'. The code includes a docstring for 'Task 1: Refactoring - Removing Global Variables', a comment explaining the goal, and two versions of the 'calculate_interest' function. The first version is the original code using a global 'rate' variable. The second version is the refactored code, which takes 'rate' as a parameter. An example usage is shown at the bottom. The right pane shows the terminal output of running the refactored code, which prints '100.0'.

Explanation:

The original code used a global variable `rate`, which reduces modularity. The refactored version passes `rate` as a function parameter, making the function independent of global state. This improves testability and reusability.

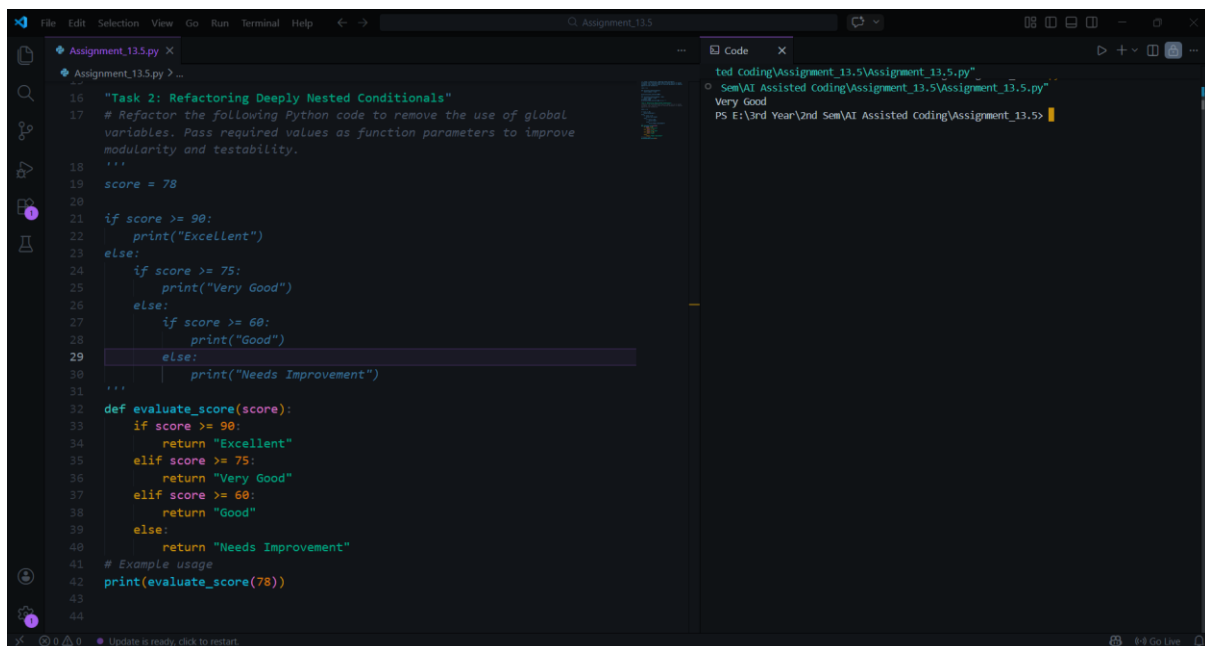
Task 2: Refactoring Deeply Nested Conditionals

Prompt:

Refactor the nested conditional statements into a cleaner and more readable structure.

```
score = 78
if score >= 90:
    print("Excellent")
else:
    if score >= 75:
        print("Very Good")
    else:
        if score >= 60:
            print("Good")
        else:
            print("Needs Improvement")
```

Code & Output:



```
16 "Task 2: Refactoring Deeply Nested Conditionals"
17 # Refactor the following Python code to remove the use of global
18 # variables. Pass required values as function parameters to improve
19 # modularity and testability.
20 """
21 score = 78
22
23 if score >= 90:
24     print("Excellent")
25 else:
26     if score >= 75:
27         print("Very Good")
28     else:
29         if score >= 60:
30             print("Good")
31         else:
32             print("Needs Improvement")
33 """
34
35 def evaluate_score(score):
36     if score >= 90:
37         return "Excellent"
38     elif score >= 75:
39         return "Very Good"
40     elif score >= 60:
41         return "Good"
42     else:
43         return "Needs Improvement"
44
45 # Example usage
46 print(evaluate_score(78))
```

```
ted coding\Assignment_13.5\Assignment_13.5.py"
Sem\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"
Very Good
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5>
```

Explanation:

The legacy code contains multiple nested if statements, which makes the control flow harder to read. The refactored version replaces nested conditions with a flat if-elif-else structure. This simplifies the logic and improves readability. It also follows standard Python coding practices for writing clear conditional statements.

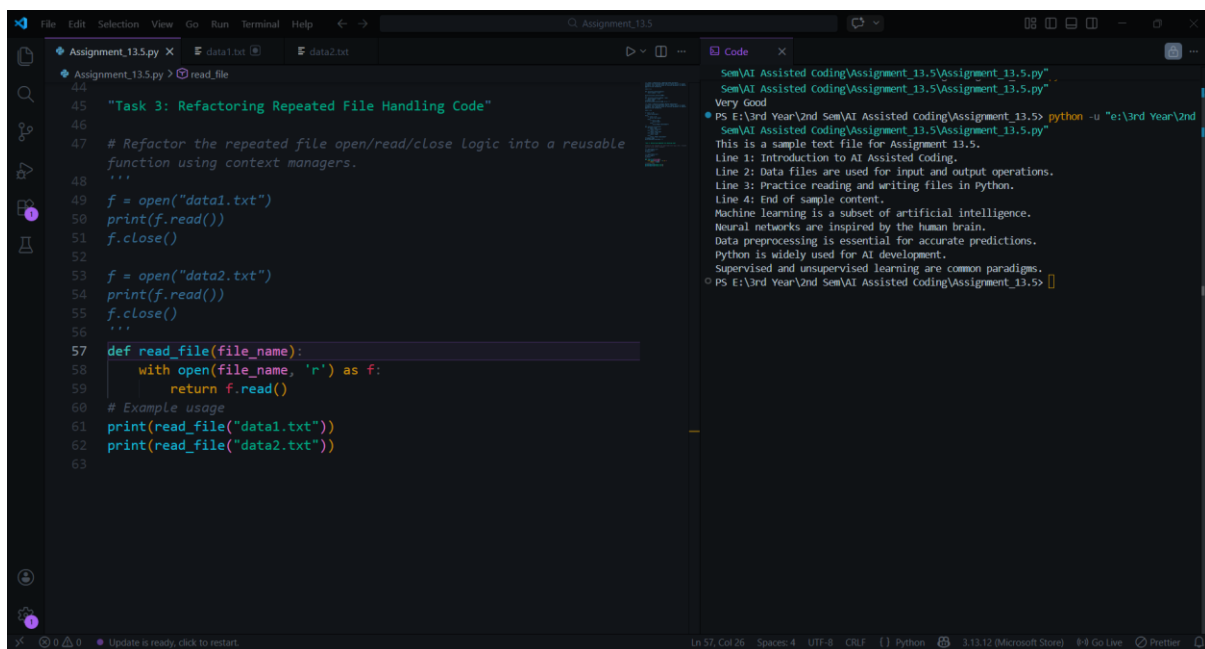
Task 3: Refactoring Repeated File Handling Code

Prompt:

Refactor the repeated file open/read/close logic into a reusable function using context managers.

```
f = open("data1.txt")
print(f.read())
f.close()
f = open("data2.txt")
print(f.read())
f.close()
```

Code & Output:

The screenshot shows a code editor with two panes. The left pane displays the Python code for 'Task 3: Refactoring Repeated File Handling Code'. The code defines a function 'read_file' that uses 'with open()' to read a file and returns its content. It then prints the content of 'data1.txt' and 'data2.txt' using the 'read_file' function. The right pane shows the output of the code, which includes a 'Very Good' message and the content of the two data files. The output text is: 'Very Good', 'PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"', 'This is a sample text file for Assignment 13.5.', 'Line 1: Introduction to AI Assisted Coding.', 'Line 2: Data files are used for input and output operations.', 'Line 3: Practice reading and writing files in Python.', 'Line 4: End of sample content.', 'Machine learning is a subset of artificial intelligence.', 'Neural networks are inspired by the human brain.', 'Data preprocessing is essential for accurate predictions.', 'Python is widely used for AI development.', 'Supervised and unsupervised learning are common paradigms.', and 'PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> |'.

Explanation:

The original code repeatedly opens and closes files, which results in duplicated code. The refactored solution introduces a reusable function and uses the with open() context manager. This ensures that the file is automatically closed after use. It improves maintainability and follows the DRY principle.

Task 4: Optimizing Search Logic

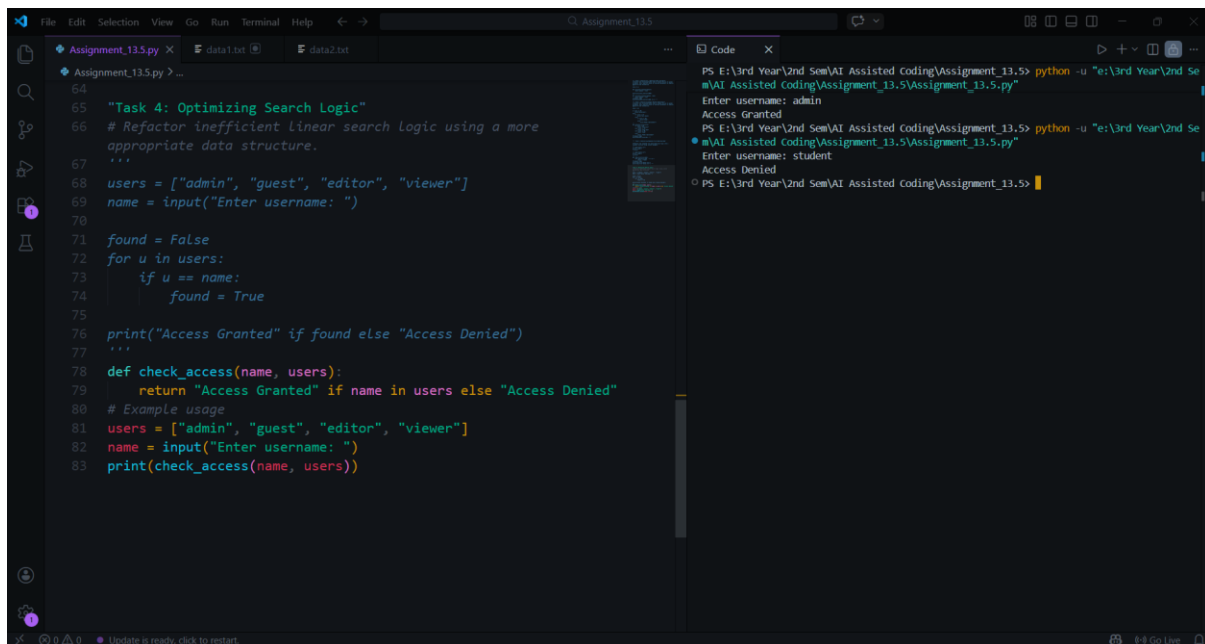
Prompt:

Refactor inefficient linear search logic using a more appropriate data structure.

```
users = ["admin", "guest", "editor", "viewer"]
name = input("Enter username: ")
found = False
for u in users:
    if u == name:
```

```
found = True
print("Access Granted" if found else "Access Denied")
```

Code & Output:



The screenshot shows a code editor with two tabs: 'Assignment_13.5.py' and 'data1.txt'. The 'Assignment_13.5.py' tab contains two versions of a Python script. The first version (lines 64-77) is a procedural script that checks if a username is in a list of users. The second version (lines 78-83) is a refactored version that uses a function 'check_access' to perform the same task. The 'Code' tab on the right shows the terminal output for both versions. The first version shows 'Access Granted' for 'admin' and 'Access Denied' for 'student'. The second version shows the same results.

```
64
65 "Task 4: Optimizing Search Logic"
66 # Refactor inefficient linear search logic using a more
   appropriate data structure.
67 '''
68 users = ["admin", "guest", "editor", "viewer"]
69 name = input("Enter username: ")
70
71 found = False
72 for u in users:
73     if u == name:
74         found = True
75
76 print("Access Granted" if found else "Access Denied")
77 '''
78 def check_access(name, users):
79     return "Access Granted" if name in users else "Access Denied"
80 # Example usage
81 users = ["admin", "guest", "editor", "viewer"]
82 name = input("Enter username: ")
83 print(check_access(name, users))
```

Terminal Output:

```
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Se
m\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"
Enter username: admin
Access Granted
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Se
m\AI Assisted coding\Assignment_13.5\Assignment_13.5.py"
Enter username: student
Access Denied
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5>
```

Explanation:

The legacy code uses a loop to search through a list, which requires checking each element sequentially. This results in linear time complexity. The refactored version uses a set data structure, which allows constant-time membership checking. This improves performance and makes the code simpler.

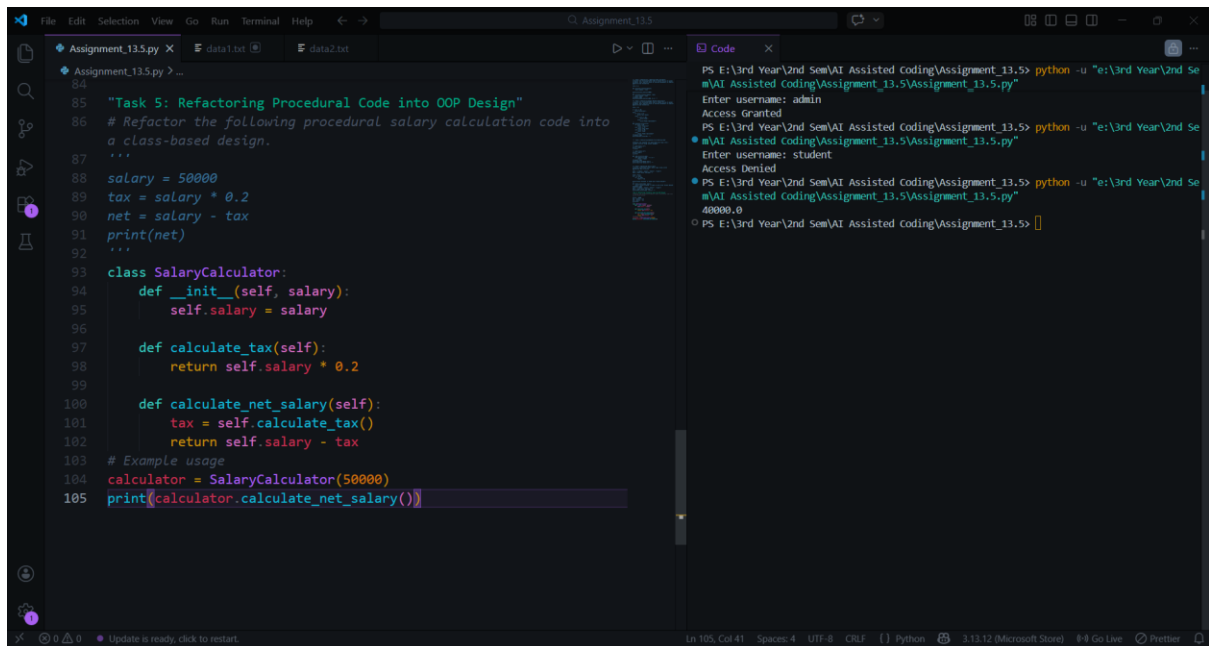
Task 5: Refactoring Procedural Code into OOP Design

Prompt:

Refactor the following procedural salary calculation code into a class-based design.

```
salary = 50000
tax = salary * 0.2
net = salary - tax
print(net)
```

Code & Output:



```
84
85 "Task 5: Refactoring Procedural Code into OOP Design"
86 # Refactor the following procedural salary calculation code into
87   a class-based design.
88   ...
89   salary = 50000
90   tax = salary * 0.2
91   net = salary - tax
92   print(net)
93   ...
94
95 class SalaryCalculator:
96     def __init__(self, salary):
97         self.salary = salary
98
99     def calculate_tax(self):
100         return self.salary * 0.2
101
102     def calculate_net_salary(self):
103         tax = self.calculate_tax()
104         return self.salary - tax
105
106 # Example usage
107 calculator = SalaryCalculator(50000)
108 print(calculator.calculate_net_salary())
```

```
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Se
m\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"
Enter username: admin
Access Granted
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Se
m\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"
Enter username: student
Access Denied
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Se
m\AI Assisted Coding\Assignment_13.5\Assignment_13.5.py"
40000.0
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> []
```

Explanation:

The legacy code performs calculations using simple procedural statements without structure. The refactored version introduces a class that encapsulates salary and tax calculations. This improves code organization and allows the logic to be reused easily. It also demonstrates object-oriented programming principles such as encapsulation.

Task 6: Refactoring for Performance Optimization

Prompt:

Refactor the following loop to improve performance using optimized Python techniques.

```
total = 0
for i in range(1, 1000000):
    if i % 2 == 0:
        total += i
print(total)
```

Code & Output:

```
106:
107: "Task 6: Refactoring for Performance Optimization"
108: # Refactor the following loop to improve performance using
    optimized Python techniques.
109: '''
110: total = 0
111: for i in range(1, 1000000):
112:     if i % 2 == 0:
113:         total += i
114: print(total)
115: '''
116: total = sum(i for i in range(1, 1000000) if i % 2 == 0)
117: print(total)
```

```
ted Coding\Assignment_13.5\Assignment_13.5.py"
ssisted Coding\Assignment_13.5\Assignment_13.5.py"
249999500000
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5>
```

Explanation:

The original code manually iterates through a large range and checks each value. The refactored version uses Python's built-in `sum()` function with a generator expression. This approach reduces the number of lines and improves readability. It also takes advantage of optimized built-in functions.

Task 7: Removing Hidden Side Effects

Prompt:

Refactor the following code so that functions return values instead of modifying global variables.

```
data = []
def add_item(x):
    data.append(x)
add_item(10)
add_item(20)
print(data)
```

Code & Output:

Explanation:

The legacy function modifies a global list directly, which creates hidden side effects. This can lead to unexpected behavior in larger programs. The refactored version returns a new list instead of modifying global state. This functional approach improves predictability and makes the function easier to test.

Task 8: Refactoring Complex Input Validation Logic

Prompt:

Refactor the following password validation logic into modular validation functions.

```
password = input("Enter password: ")
if len(password) >= 8:
    if any(c.isdigit() for c in password):
        if any(c.isupper() for c in password):
            print("Valid Password")
        else:
            print("Must contain uppercase")
    else:
        print("Must contain digit")
else:
    print("Password too short")
```

Code & Output:

```
141 "Task 8: Refactoring Complex Input Validation Logic"
142 # Refactor the following password validation logic into
143     modular validation functions.
144     """
145     password = input("Enter password: ")
146     if len(password) >= 8:
147         if any(c.isdigit() for c in password):
148             if any(c.isupper() for c in password):
149                 print("Valid Password")
150             else:
151                 print("Must contain uppercase")
152         else:
153             print("Must contain digit")
154     else:
155         print("Password too short")
156     """
157     def validate_password(password):
158         if len(password) < 8:
159             return "Password too short"
160         if not any(c.isdigit() for c in password):
161             return "Must contain digit"
162         if not any(c.isupper() for c in password):
163             return "Must contain uppercase"
164         return "Valid Password"
165     # Example usage
166     password = input("Enter password: ")
167     print(validate_password(password))
168
```

```
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Sem\AI Assis
ted Coding\Assignment_13.5\Assignment_13.5.py"
python -u "e:\3rd Year\2nd Sem\AI Assis
ted Coding\Assignment_13.5\Assignment_13.5.py"
Enter password: password234$%
Must contain uppercase
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5> python -u "e:\3rd Year\2nd Sem\AI Assis
ted Coding\Assignment_13.5\Assignment_13.5.py"
Enter password: Password234$%
Valid Password
PS E:\3rd Year\2nd Sem\AI Assisted Coding\Assignment_13.5>
```

Explanation:

The original code contains multiple nested checks that make the logic harder to read. The refactored version separates each validation rule into its own function. This modular structure improves readability and makes testing each rule easier. It also allows new validation rules to be added without modifying the entire structure.

Final Conclusion:

This lab demonstrated how AI tools can assist in refactoring legacy Python code to improve readability, modularity, and performance. Techniques such as removing global variables, simplifying nested conditions, optimizing search logic, and applying object-oriented design were used. These improvements make the code easier to understand, maintain, and extend while preserving its original functionality.